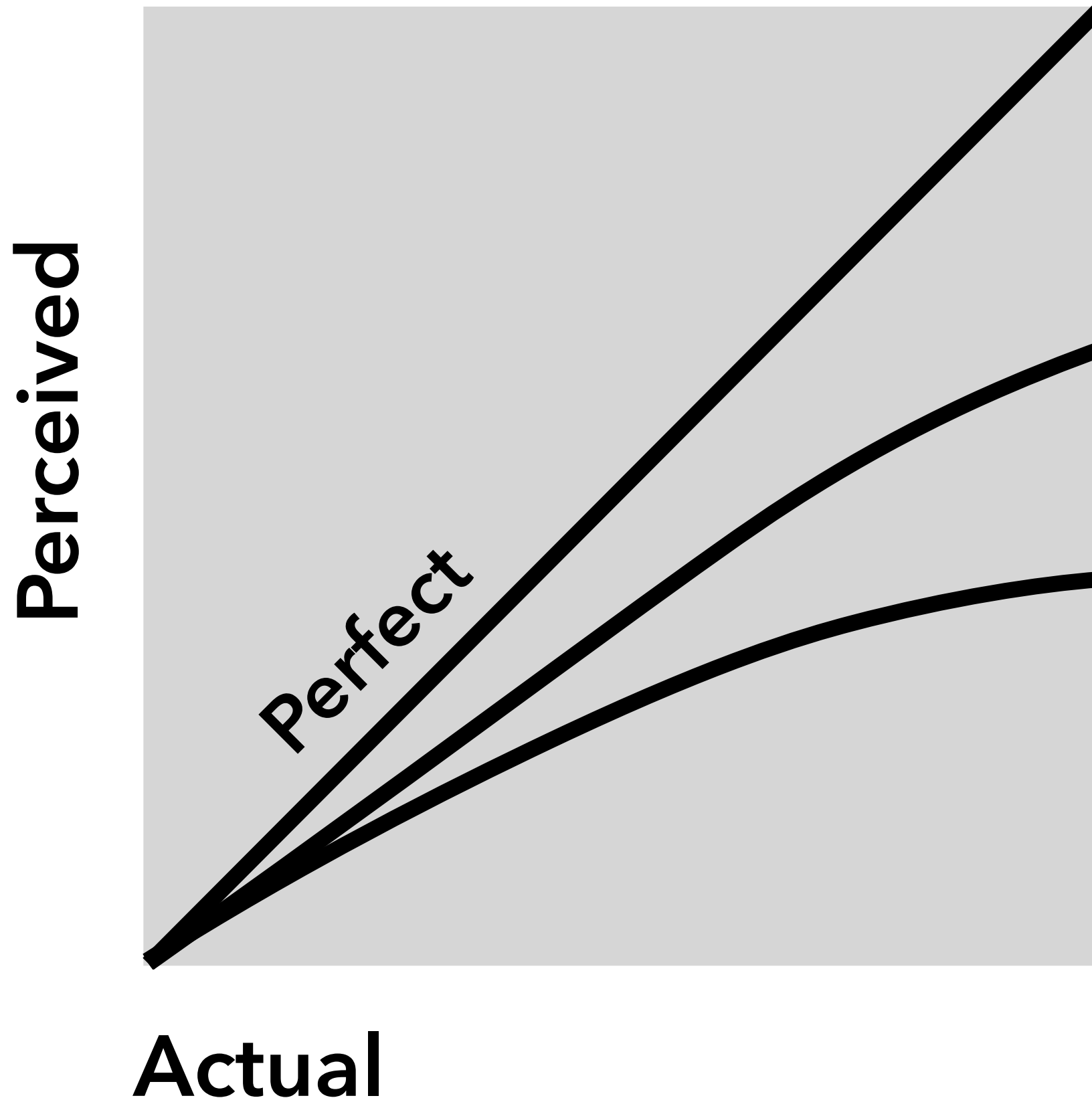


Stevens' power law

$$\Phi(I) = kI^a$$



Length

of symbol estimated
correctly



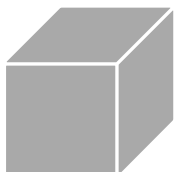
Area

of symbol
underestimated



Volume

of symbol
underestimated



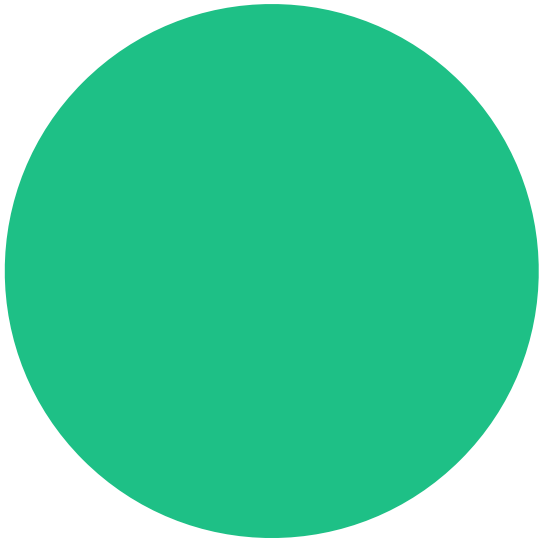
Encoding induces bias

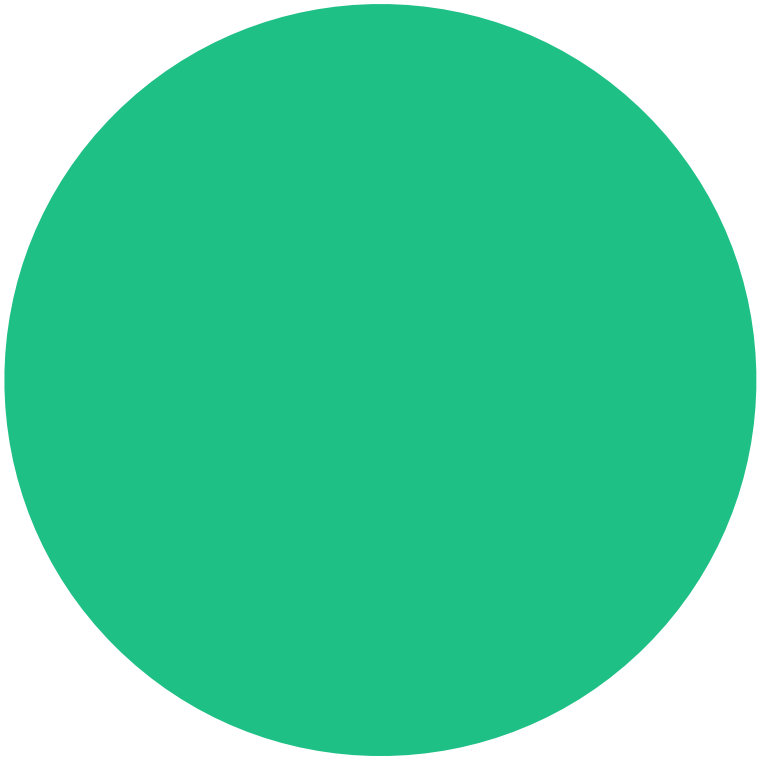
2

0









400

600

200

$$2822 = 2000 * \sqrt{2}$$

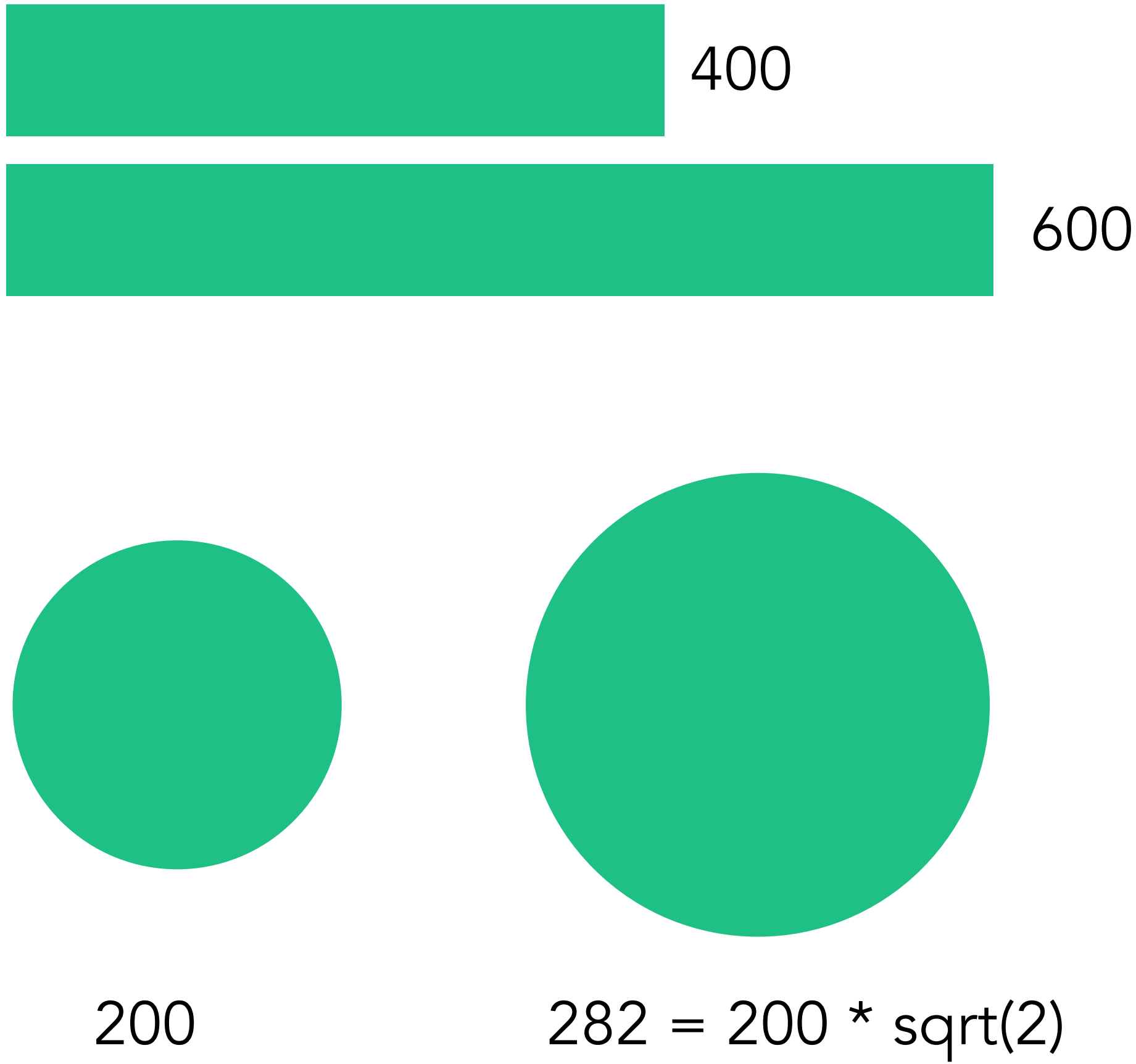


underestimate



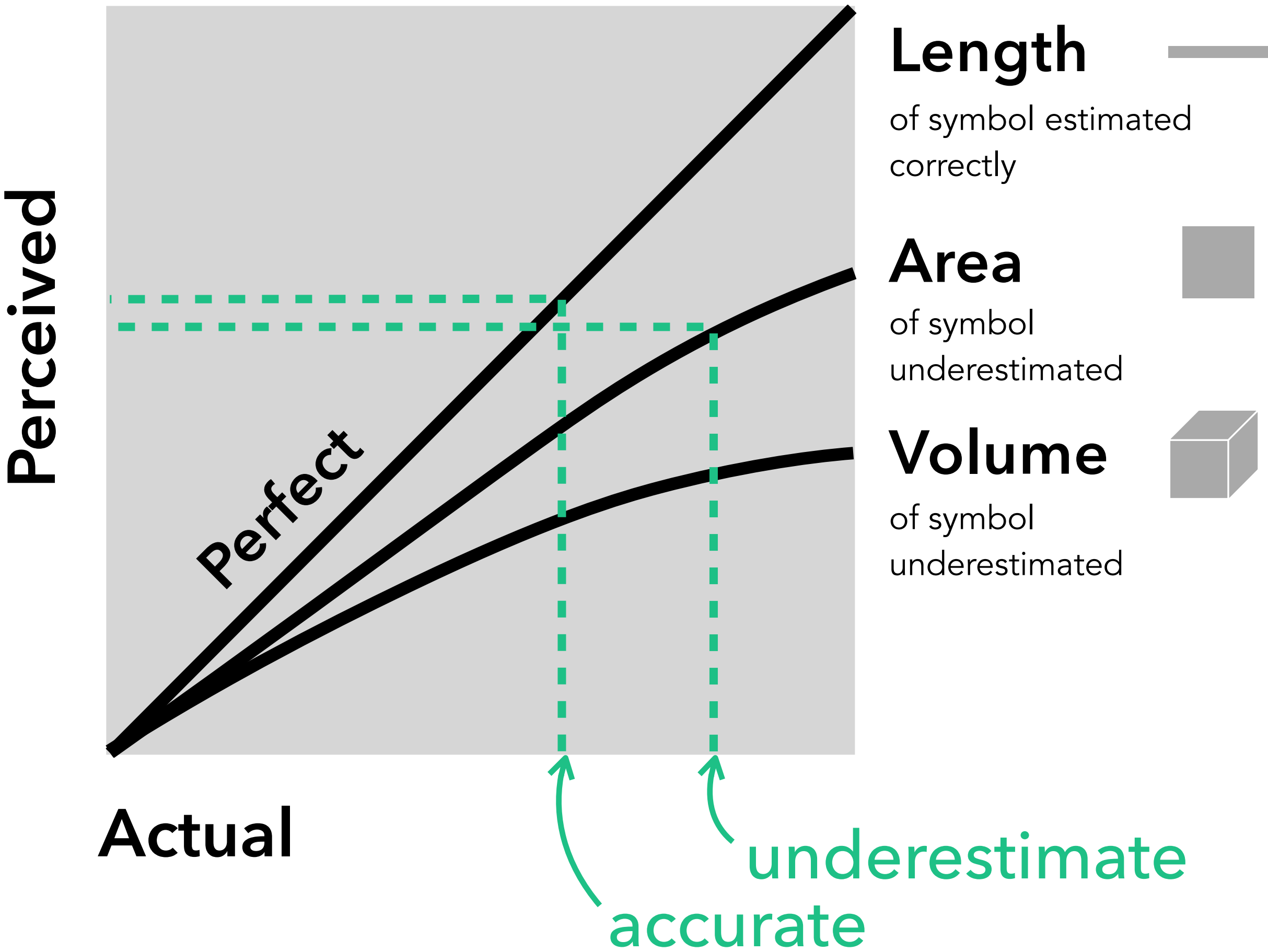
accurate

Encodings induce bias



Stevens' power law

$$\Phi(I) = kI^a$$



Continuum	a	Stimulus condition
Brightness	0.5	Point source
Smell	0.6	Heptane
Loudness	0.67	Sound pressure of 3000 Hz tone
Visual area	0.7	Projected square
Warmth	0.7	Irradiation of skin, large area
Taste	0.8	Saccharin
Vibration	0.95	Amplitude of 60 Hz on finger
Brightness	1	Point source briefly flashed
Cold	1	Metal contact on arm
Thermal pain	1	Radiant heat on skin
Visual length	1	Projected line
Duration	1.1	White-noise stimuli
Vocal effort	1.1	Vocal sound pressure
Lightness	1.2	Reflectance of gray papers
Taste	1.3	Sucrose
Warmth	1.3	Irradiation of skin, small area
Electric shock	3.5	Current through fingers

Stevens' power law

$$\Phi(I) = kI^a$$